Haley A. Oleynik

PhD Candidate
Institute for the Oceans and Fisheries, University of British Columbia
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EDUCATION

Colby College, *Waterville, ME*, **B.A.** Biology & Sociology 2015

University of Delaware, *Lewes, DE*, **M.S.** Marine Science 2020

University of British Columbia, *Vancouver, BC*, **PhD** Fisheries Science in progress

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RESEARCH EXPERIENCE

PhD Candidate Sept 2021 - Present

Institute for the Oceans and Fisheries University of British Columbia

My PhD research involves population modeling of 28 stocks from five species of Fraser River salmon and steelhead (*Oncorhynchus spp.*) to identify ecological drivers of productivity declines, with the goal of understanding regional, multi-species population patterns and incorporating ecosystem information into fisheries and ecosystem management.
Skills: Statistical modeling, Bayesian modeling, R programming, Git/Github, stakeholder engagement

Graduate Research Assistant Jan 2024 - Present

United States Army Corps of Engineers

As a part-time research assistant I conduct statistical modeling, using Bayesian mark-recapture models in JAGS, zero-inflated binomial regressions, and generalized additive models, to determine Chinook salmon (*Oncorhynchus tshawytscha*) passage success and survival through dams in the Willamette River system in Oregon.
Skills: Statistical modeling, Bayesian modeling, JAGS programming, data analysis, stakeholder engagement, R programming

Research Analyst Feb 2026 - May 2026

Pacific Salmon Commission Vancouver, B.C.

During a three month contract I analyzed salmon monitoring data to statistically evaluate the use of a new monitoring method for salmon passage through Hells Gate, a narrow gorge which poses passage challenges for sockeye (*Oncorhynchus nerka*) and pink salmon (*Oncorhynchus gorbuscha*) migrating up the Fraser River to their spawning grounds.
Skills: Statistical modeling, R programming, Git/Github

Aquatic Biologist II Jun 2025 - March 2026

Department of Fisheries and Oceans Canada Vancouver, B.C.

As a contract research biologist, I used used empirical harbour seal diet data and modeled diet composition from Ecopath with Ecospace (EwE) ecosystem models to statistically validate model outputs and inform field sampling programs.
Skills: Spatio-temporal modeling, VAST, ecosystem modeling, EwE, bioenergetics modeling, TMB programming

M.S. Student Sept 2018 - July 2020

School of Marine Science and Policy University of Delaware

My Master's research involved analyzing data from a state-run trawl survey using multivariate statistical methods to describe the fish and invertebrate community in Delaware Bay, identify physical drivers of community composition, and evaluate long-term changes in the community.
Skills: Fishery independent data analysis, multivariate statistics, timeseries analysis, R programming

Undergraduate Senior Honors Thesis Sept 2014 - May 2015

Department of Sociology Colby College

My undergraduate honors thesis project involved conducting in-depth interviews with harvesters to understand social and community implications of the changing American lobster (*Homarus americanus*) population, and subsequent management, on the lobster fishing industry in Maine.
Skills: Sociological research, interview design, stakeholder engagement

Research Intern June 2014 - Aug 2014

CIEE Bonaire Bonaire, Netherland Antilles

I used SCUBA field sampling methods to catch invasive lionfish, *Pterois*, and dissect and analyze stomach contents, to explore the predatory impact lionfish have on juvenile fish in Bonaire.
Skills: SCUBA, stomach content analysis, field sampling

Research Intern Jan 2013 - Feb 2013

Smithsonian Natural History Museum Washington D.C.

I curated, identified, and photographed over 50 species of *pteropoda* to catalog morphology for research of the genetics, morphology, and biogeographic information of pteropods to better understand ocean connectivity.
Skills: Curation, phylogeny, Geneious software, Image J software

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PROFESSIONAL EXPERIENCE

Teaching Assistant Sept 2023 - Present

Master of Data Science Program University of British Columbia

For four years, I have been a part-time teaching assistant for several MDS statistics and computation courses. I lead coding labs, evaluate student work, and provide feedback. The courses included, *Programming for Data Manipulation* (DSCI 523), *Statistical Inference and Computation I* (DSCI 552), *Regression I* (DSCI 561), *Regression II* (DSCI 562), *Statistical Inference and Computation II* (DSCI 553), *Experimentation and Causal Inference* (DSCI 554).

Teaching Assistant Spring 2026

Ecosystem Modeling with EwE University of British Columbia

As a teaching assistant for a graduate-level Ecosystem modeling course (FISH 501), which taught the use of Ecopath with Ecosim (EwE) for marine ecosystem modeling, I delivered lectures and provided feedback to students on course projects.

Graduate Writing Consultant Sept 2022 - April 2024

Center for Writing and Scholarly Communication University of British Columbia

As a writing consultant I provided written and in-person consultations to graduate and undergraduate students on written work including class assignments, manuscripts, and dissertations.

NOAA Sea Grant John A. Knauss Marine Policy Fellow Feb 2021 - Feb 2022

National Stock Assessment Program NOAA Fisheries

As a Knauss Fellow I supported the National Stock Assessment Program by organizing a technical community, developed the Fisheries Integrated Toolbox, assisted in coordination of a new national stock assessment modeling framework (FIMS), and facilitated regional communication, to advance the NOAA stock assessment enterprise.

Communications Committee Chair Oct 2022 - June 2026

Committee Member June 2015 - Oct 2022

Association of the Marine Laboratories of the Caribbean

In this volunteer position I managed internal and external communication, and helped organize three biennial scientific conferences, for the AMLC, a nonprofit which connects marine laboratories, scientists, students, and stakeholders in the Wider Caribbean Region.

North Pacific Groundfish Observer June 2016 – Jan 2018

Saltwater Inc. / NOAA Fisheries

As a fisheries observer I collected data on catch, bycatch, and fishing effort on board commercial Alaskan fishing vessels to be used by the National Marine Fisheries Service to monitor fish populations, set catch limits, and manage commercial Alaskan fisheries.

PUBLICATIONS

Brodziak, J., M. Supernaw, B. Li, C. Stawitz, **H.A. Oleynik**, T. A'mar. 2025. Engineering a prototype for a next generation stock assessment modeling platform. Fishery Bulletin. 123(2), 88-104.

Oleynik, H.A., J.J Bizzarro, E.A. Hale, A.B. Carlisle. 2024. Environmental drivers of biogeography and community structure in a Mid-Atlantic estuary. Oecologia. 204(3), 543-557.

Zahner. J.A, M.A. Heller-Shipley, **H.A. Oleynik**, S.G. Beyer, P-Y. Hernvann, M. Véron, A.N. Odell, J.Y. Sullivan, A.L. Hayes, V.G. Gertseva, K.L. Oken, O.S. Hamel, M.A. Haltuch. 2023. *Status of Shortspine Thornyhead (Sebastolobus alascanus) along the US West coast in 2023*. Pacific Fishery Management Council, Portland, Oregon. 139 p.

Blackhart, K., and **H.A. Oleynik**. 2023. *Setting targets and analyzing data gaps for U.S. fish assessments*. NOAA Tech. Memo. NMFS-F/SPO-242, p. 77

Oleynik, H.A. 2022. *Living Resources: Fish Population Trends in Delware Bay: Climate and Community*. Technical Report for the Delaware Estuary and Basin, J. Shinn, L. Haaf, L. Morgan, and D. Kreeger (eds). Partnership for the Delaware Estuary.

ICES (2022): Working Group on Technology Integration for Fishery-Dependent Data (WGTIFD; outputs from 2021 meeting). ICES Scientific Reports. Report.

Mathews, J.A., E.A. Hale, **H.A. Oleynik**, et al. (2022) *Exploring Trends in Abundance of Young-of-the-Year and Age-1 Atlantic Croaker, Black Drum, Spot, and Weakfish in Relation to Salinity, Temperature, and Large-Scale Climatic Signals in a Mid-Atlantic Estuary.* Transactions of the American Fisheries Society, 151(2), 150–171.

OUTREACH AND ENGAGEMENT

REDI Committee **Student rep**, UBC Institute for the Oceans and Fisheries 2022-2025

Student Society **Social activities chair**, UBC Institute for the Oceans and Fisheries 2022-2024

Society for Women in Marine Science **Co-Chair**, University of Delaware Chapter 2019-2020

Graduate Student Association **Co-Chair**, UD School of Marine Science and Policy 2019-2020

Hutton Junior Fisheries Biology Internship **Reviewer**, American Fisheries Society 2024

John A. Knauss Marine Policy Fellowship **Reviewer**, NOAA Sea Grant 2022, 2024

EXPERTISE

Computing Advanced in R, WinBUGS, JAGS; experience with C++, Stan, Stock Synthesis, Template Model Builder (TMB), Vector Autoregressive Spatio-Temporal modeling (VAST)

Field & Lab AAUS Scientific Diver, SDI Rescue Diver, small boat operation, random field sampling, otolith collection, shark tagging, marine animal husbandry

AWARDS AND FUNDING

Partnership Engage Grant ($23,000) *Social Sciences and Humanities Research Council, Canada* 2024

Buck Suzuki Legacy Bursary Award ($1,000) *T. Buck Suzuki Foundation* 2022

Frances Severance Thesis Distinction Award *University of Delaware* 2021

Daiber Fellowship ($27,000) *University of Delaware* 2019

Women of Promise Award *University of Delaware* 2019

Okie Fellowship ($26,000) *University of Delaware* 2018

PRESENTATIONS & GUEST LECTURES

"Fisheries Oceanography" Guest Lecture, **Biological Oceanography Summer Course**, Bamfield Marine Science Center, Bamfield, BC 2026

"Impacts of Dam Operations on Mortality and Injury of Juvenile Chinook Salmon at Cougar Dam" **Willamette Science Review**, Corvallis, OR 2026

"Cougar Dam Gate Opening Analysis: Impacts on Chinook Mortality" **Willamette Valley System Adaptive Management Workshop**, Salem, OR 2026

"Population-level survival impacts of Harbour seal foraging on juvenile Pacific salmon, and the mechanistic links" **PICES**, Yokohama, Japan 2025

"Drivers of Productivity in Fraser River Salmon" **American Fisheries Society (AFS) WA-BC Chapter Meeting**, Vancouver, BC 2025

"Incorperating Human Dimensions into EBFM: Understanding Socioeconomic Consequences of Ecological Change" **American Fisheries Society Annual Conference**, Honolulu, HI 2024

"Integrating Electronic Monitoring Data into U.S. Fish Stock Assessments" **World Fisheries Congress**, Seattle, WA 2024

"Fisheries Science and Management" Guest Lecture, **George Washington University Marine Ecology Course** (BISC 3454), Washington DC 2024

"Long-Term Shifts in the Fish and Macroinvertebrate Community in a Mid-Atlantic Estuary" **American Fisheries Society Annual Conference**, remote 2021

"Changes in a Mid Atlantic Estuary: Long Term Shifts in the Fish and Macroinvertebrate Community in Delaware Bay" **NOAA Central Library Seminar Series**, remote 2021

"Long-Term Shifts in the Fish and Macroinvertebrate Community in a North American Estuary" **World Fisheries Congress, remote** 2021

"Changes in a Mid-Atlantic Estuary: Long-Term Trends in Environmental Conditions and Marine Community in Delaware Bay" **AFS Mid-Atlantic Chapter Meeting**, Lewes, DE 2020